

Arpit Kapoor

Postdoctoral Researcher in Machine Learning and Hydrology

Phone: +61 412 982 005 | Email: kapoor.arpit97@gmail.com

Address: 4 Matilda Way, Glenwood, NSW 2768, Australia

www.arpit-kapoor.com | [GitHub \(arpit-kapoor\)](https://github.com/arpit-kapoor) | [Google Scholar](https://scholar.google.com/citations?user=...) | [LinkedIn](https://www.linkedin.com/in/arpitkapoor/)

Research Summary

Postdoctoral Research Associate at the University of Sydney with PhD in Mathematics & Statistics from UNSW, specialising in Machine Learning for Water Resources Modelling. My research develops hybrid and Bayesian deep learning approaches that integrate physical models with machine learning to advance rainfall-runoff modelling, flood forecasting, and groundwater flow emulation, with publications in leading journals. I have expertise in Bayesian inference, machine learning, uncertainty quantification, and high-performance computing, alongside experience with large-scale environmental and remote sensing datasets. Beyond academia, I worked as a Climate Science Support Officer at the Bureau of Meteorology and in data science roles, developing scalable ML pipelines, climate model bias correction, and reproducible workflows. I aim to create innovative, interpretable models that improve prediction and decision-making in water resources while advancing the state of scientific ML.

Education

Doctor of Philosophy (PhD), Mathematics and Statistics

Aug 2022 – April 2026

University of New South Wales, Sydney

Grants: UNSW University International Postgraduate Award (UIPA)

Bachelor of Technology (B.Tech)

Jul 2015 – May 2019

SRM Institute of Science and Technology, India

Major: Computer Science & Engineering (GPA: 9.01/10.0)

Final Project: Deep reinforcement learning methods for locomotion of a humanoid robot

Professional Experience

The University of Sydney

Mar 2026 – Present

Postdoctoral Research Associate, Faculty of Engineering

- Research focus on development of novel approaches synergising Physics-based hydrologic and groundwater models with state-of-the-art machine learning approaches
- Facilitate the organisation of Sydney Hydrology Day 2027
- Developing Data Science suite for Spatiotemporal Water Quality Analysis for Hunter Water Corporation

Bureau of Meteorology, Australia

Feb 2023 – Apr 2025

Climate Science Support Officer

- Implemented and evaluated a **multivariate bias correction method (MRNBC)** for climate projections to support the **Australian Climate Service**, improving the reliability of projections for downstream tasks of hydrological modelling.

- Developed a modern **Python interface for legacy FORTRAN software**, enabling integration with current scientific workflows and facilitating reproducibility
- Applied distributed computing with **Dask on the National Computing Infrastructure (NCI)**, **reducing model runtimes by 80%**.
- Co-authored an **abstract for EGU 2025** on bias adjustment techniques for climate data and contributed to a **forthcoming Bureau research report** on MRNBC implementation

Quince, India

Mar 2022 – Aug 2022

Data Scientist

- Applied **self-supervised learning** techniques to model and predict customer churn, which informed strategies leading to a **15% lift in user engagement**.
- Developed a **gradient boosting-based forecasting model** (logistics), highlighting skills in model design, optimisation, and validation for large-scale applied datasets.
- Worked in an interdisciplinary setting, integrating machine learning outputs into decision systems — experience directly transferable to applied environmental modelling pipelines.

3Qi Labs, India

Nov 2019 – Nov 2021

Data Scientist

- Designed and deployed an LSTM AutoEncoder for **anomaly detection** in high-volume data pipelines, demonstrating **expertise in sequence modelling and evaluation of unsupervised methods**.
- This approach led to a **40% uplift in the detection accuracy** compared to the existing approach
- Built **automated machine learning** workflows using distributed computing platforms (PySpark & Databricks), developing reproducible pipelines **suited to large-scale scientific data**.
- Introduced and deployed **MLOps infrastructure** (MLflow & Docker), improving model lifecycle management.

BoMoTix, India

Jan 2019 – Nov 2019

Machine Learning Engineer

- Translate state-of-the-art research in **computer vision** to applications in sports analysis, bridging academic methods with real-world applications.
- Designed **deep learning** based solutions for **computer vision** problems like player **detection and tracking** (based on SiameseNet and DeepSORT), 3D player **pose estimation** (based on DeepHAR)
- Improved model efficiency using **mixed-precision and GPU optimisations**, leading to a 15% reduction in model training costs
- Cleaned, processed, and curated video data for model training and validation
- Led the effort on methodological documentation and authored technical **white papers**.

The University of Sydney

Jan 2018 – Sep 2018

Visiting Student Researcher

- Contribute to research on **Bayesian methods** for uncertainty quantification in neural networks
- Co-developed a parallel **Markov chain Monte-Carlo (MCMC)** based approach for **Bayesian transfer learning in neural networks**
- Provided programming support for the development of a **Bayesian inversion** approach in **geoscientific models** using parallel-tempering MCMC methods
- Co-authored **publications on Bayesian machine learning** at top-tier ML journals

Core Projects

Enhancing environmental process modelling with scientific deep learning

- **Variational RNNs for Cyclone behaviour prediction** – Applying variational recurrent neural networks to forecast cyclone trajectory and intensity with probabilistic confidence estimates.
- **DeepGR4J: Hybrid rainfall-runoff modelling** – Designing a deep learning hybridization of the GR4J conceptual hydrological model to advance streamflow prediction.
- **Quantile-based ensemble deep learning for flood forecasting** – Building a quantile regression-based ensemble framework for streamflow and flood prediction, with emphasis on uncertainty quantification.
- **QDeepGR4J: quantile-based ensemble of hybrid model** – Extending hybrid deep learning models with quantile regression to better capture extremes and propagate uncertainty.
- **Neural operators for groundwater modelling** – Developing geometry-informed neural operator (GINO) approaches to emulate 3D groundwater flow in coastal aquifers with transient conditions and variable density.

Grants & Awards

- PhD Scholarship, Australian Research Council Training Centre in Data Analytics for Resources and Environment (DARE Centre)
- PhD Scholarship, UNSW University International Postgraduate Award (UIPA)
- Finalist at the IEEE/RSJ IROS 2017 Humanoid Application Challenge (Canada)
- Secured 1 Gold, 2 Silver, 1 Bronze medal at Robogames 2017 (USA)
- Recipient of travel grant to participate in research at the Centre for Translational Data Science (CTDS), University of Sydney (2018)

Academic Activities

- Contribute to Machine Learning tutorial development for ENVX3003: Hydrologic Monitoring and Modelling course in T2 2026 at University of Sydney
- Develop and maintain the official website of the Sydney Hydrology Day. Continue to support the organisation of the annual event connecting hydrologists in the Sydney basin area.
- Co-organiser, MODSIM 2025, Session “C7. Advancing Machine Learning in Hydrology: Ensuring Reliable, Reproducible, and Validated Applications”
- Presented a 10-minute thesis at the DARE Symposium hosted by the ARC training centre for Data Analytics for Resources and Environment (DARE Centre) (2025) (video link: <https://www.youtube.com/watch?v=3zhcXW1qfXQ>)
- Selected Participant, Information Resilience PhD School, University of Queensland (2024)
- Challenge Facilitator, Data Study Group, Alan Turing Institute (UK, May 2024)
- Invited for talk on DeepGR4J project at the HydroCafe, The Bureau of Meteorology (2024)

- Delivered a talk on scientific machine learning at the Postgraduate Conference, School of Mathematics and Statistics, University of New South Wales, Sydney (2024)
- Poster presentation on hybrid hydrological modelling at the Postgraduate Conference, School of Mathematics and Statistics, University of New South Wales, Sydney (2023)
- Presented at UNSW Transitional AI research group seminar series (video link: <https://www.youtube.com/watch?v=F9Gi8xVlpQU>, <https://www.youtube.com/watch?v=QR2xoyzNKyo>)
- Co-organised and hosted the UNSW Transitional AI research group seminar series (2023-2024)
- Tutored for postgraduate machine learning courses at UNSW (MATH3856, MATH5836 and COMP9417)
- Affiliate student researcher at the UNSW AI Institute (UNSW Data Science Hub) (2023 - Present)
- Team Leader - SRM Team Humanoid, student-led humanoid robotics research (2018-2019)

Publications

2026-Present

Irving, D., Takbash, A., Peter, J., Gammon, A., Dowdy, A., Dao, T.L., **Kapoor, A.**, Black, M., Jakob, D. and Grose, M., 2026. An intercomparison of climate model bias correction methods across Australia. *Journal of Southern Hemisphere Earth Systems Science*, 76(2), p.ES25023.

2023-2025

Kapoor, A., & Chandra, R. (2025). QDeepGR4J: Quantile-based ensemble of deep learning and GR4J hybrid rainfall-runoff models for extreme flow prediction with uncertainty quantification. *Journal of Hydrology*, 134434.

Takbash, A., Irving, D., Peter, J., Dao, T. L., **Kapoor, A.**, Gammon, A., Dowdy, A., Black, M., Bende-Michl, U., Jakob, D., and Grose, M.: A Comprehensive Assessment of Climate Data Bias-Adjustment Techniques Over Australia, EGU General Assembly 2025, Vienna, Austria, 27 Apr–2 May 2025, EGU25-252

Chandra, R., **Kapoor, A.**, Khedkar, S., Ng, J., & Vervoort, R. W. (2024). Ensemble quantile-based deep learning framework for streamflow and flood prediction in Australian catchments. *arXiv preprint arXiv:2407.15882*.

Kapoor, A., Pathiraja, S., Marshall, L., & Chandra, R. (2023). DeepGR4J: A deep learning hybridization approach for conceptual rainfall-runoff modelling. *Environmental Modelling & Software*, 169, 105831.

Kapoor, A., Negi, A., Marshall, L., & Chandra, R. (2023). Cyclone trajectory and intensity prediction with uncertainty quantification using variational recurrent neural networks. *Environmental Modelling & Software*, 162, 105654.

2020 - 2022

Kapoor, A., Nukala, E., & Chandra, R. (2022). Bayesian neuroevolution using distributed swarm optimization and tempered MCMC. *Applied Soft Computing*, 129, 109528.

Chandra, R., Azam, D., **Kapoor, A.**, & Müller, R. D. (2020). Surrogate-assisted Bayesian inversion for landscape and basin evolution models. *Geoscientific Model Development*, 13(7), 2959-2979.

Chandra, R., Jain, K., **Kapoor, A.**, & Aman, A. (2020). Surrogate-assisted parallel tempering for Bayesian neural learning. *Engineering Applications of Artificial Intelligence*, 94, 103700.

Chandra, R., & **Kapoor, A.** (2020). Bayesian neural multi-source transfer learning. *Neurocomputing*, 378, 54-64.

2017 - 2019

Sripada, A., Asokan, H., Warriar, A., **Kapoor, A.**, Gaur, H., Patel, R., & Sridhar, R. (2018). Teleoperation of a humanoid robot with motion imitation and legged locomotion. In *2018 3rd International Conference on Advanced Robotics and Mechatronics (ICARM)* (pp. 375-379). IEEE.

Sripada, A., Warriar, A., **Kapoor, A.**, Gaur, H., & Hemalatha, B. (2017). Dynamic lateral balance of humanoid robots on unstable surfaces. In *2017 International Conference on Electrical, Electronics, Communication, Computer, and Optimization Techniques (ICEECCOT)* (pp. 1-6). IEEE.